

FileSonic disables file sharing

In wake of the recent crackdown on Megaupload, FileSonic has disabled its file sharing service
<http://bit.ly/y20koh>

Google+ hits 90 million users

Google's recent sign-up spree has helped the social network achieve 90 million users.
<http://bit.ly/yp21qv>

Voice Modulator

Tenor to baritone with the flick of a switch

Mithun Mohandas
readersletters@thinkdigit.com

Ever fancied playing pranks on your unsuspecting friends? Of course you did, and this DIY here is one you should have in your arsenal of tricks.

Stuff that can be accomplished using this.

- Phone pranks by changing your own voice (Don't do anything illegal!)
- Use it as an inline audio modulator on your friend's sound system. (You are responsible if you blow his system)
- Make a creepy recording in an alien voice and send the tabloids in a frenzy. (Remember the aliens drinking cow's milk story on a news channel?)
- Wear a scary costume and modulate your voice.
- Start a singing career!
- ... and so much more.

What we'll be building this month is a Voice modulator and the table alongside lists what all you will need to make that happen.

Construction

The construction of the circuit is fairly simple and we've even included one CAD

drawing if you want to etch your own circuit (<http://bit.ly/Ab1rfG>). The circuit uses a Dual Op-amp (LM358) and an amplifier TDA7052 to get the job done. One amp of LM358 acts as a preamplifier for the microphone, while the other amp acts as a Wein bridge oscillator. In layman's terms, one half increases the signal strength of the mic and the other modifies the signal. The other IC TDA7052 is an audio amplifier which will drive the 8 Ω (Ohm) speaker. You may connect the speaker, mic and a voltage source in serial to see if your voice is picked up by the mic and audible via the speaker. Most probably, it won't. That's why the amplifier is needed.

You can modify the circuit according to your needs. Changing the op-amp will provide for different sound signatures. The resistance of the two trimmers (R3 & R9) can be varied to produce different sound signatures also since it modifies the frequency of the Wein bridge. If you notice too much noise emanating from the speaker then swap mics or insert an audio jack at the mic's location. This way you can modify a song's playback.

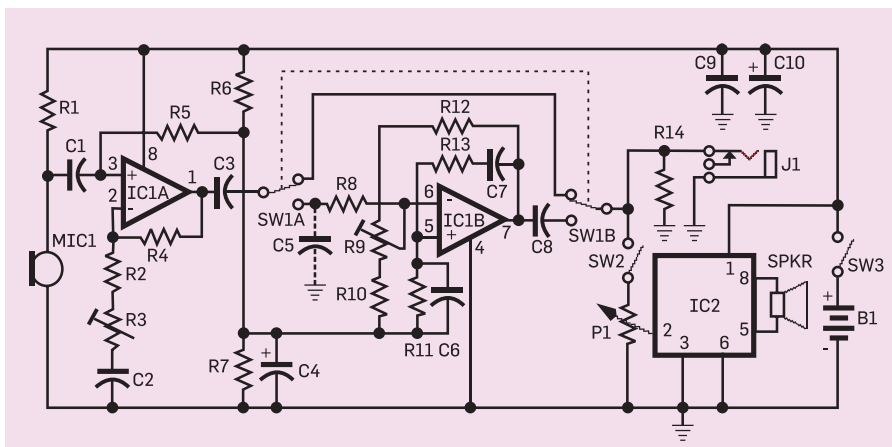
The switch SW1 provides for a way to bypass the voice modulation part of the circuit, so you can provide a clean signal as and

when required. So with a flick of this switch the voice changes altogether.

You may replace the microphones with an audio jack which will enable you to use better microphones and get a cleaner audio signal. Ensure that the capacitor

Component list

Parts	Value	Particulars
P1	10 K Ω	Log Potentiometer
R1, R10	10 K Ω 1/4W	Resistor
R2	1 K Ω 1/4W	Resistor
R3	50 K Ω 1/2W	Trimmer
R4, R6, R7, R14	100 K Ω 1/4W	Resistor
R5	47 K Ω 1/4W	Resistor
R8	68 K Ω 1/4W	Resistor
R9	2.2 K Ω 1/2W	Trimmer
R11	33 K Ω 1/4W	Resistor
R12	18 K Ω 1/4W	Resistor
R13	15 K Ω 1/4W	Resistor
C1, C2, C3, C8, C9	100 nF/63 V	Polyester Capacitor
C4	10 μ F/25 V	Electrolytic Capacitor
C5	220 nF/63 V	Polyester Capacitor
C6	4.7 nF/63 V	Polyester Capacitor
C7	10 nF/63 V	Polyester Capacitor
C10	220 μ F/25 V	Electrolytic Capacitor
IC1	LM358	Low Power Dual Op-amp
IC2	TDA7052	Audio power amplifier
MIC1		Electret Condensor mic
SPKR	8 Ω	Loudspeaker
SW1	DPDT	Toggle switch
SW2, SW3	SPST	Toggle switch
J1	3.5mm	Mono audio jack
B1	9V	DC battery



Circuit Diagram

leads are short and have a snug fit to reduce unwanted noise from creeping into the signal due to inductance. The same applies for all components when an audio circuit is to be assembled. So ensure all your components are of good quality or you'll be forced to bear with a persistent buzzing sound. Stick to polymer capacitors as they have a lower ESR than electrolytic capacitors. Condenser microphones cannot be helped so bypassing that with an audio jack is your only option. Also, potentiometers tend to be of poor quality and have a high tolerance, so you may require a shunt or series resistor to bring the value to 10K.